

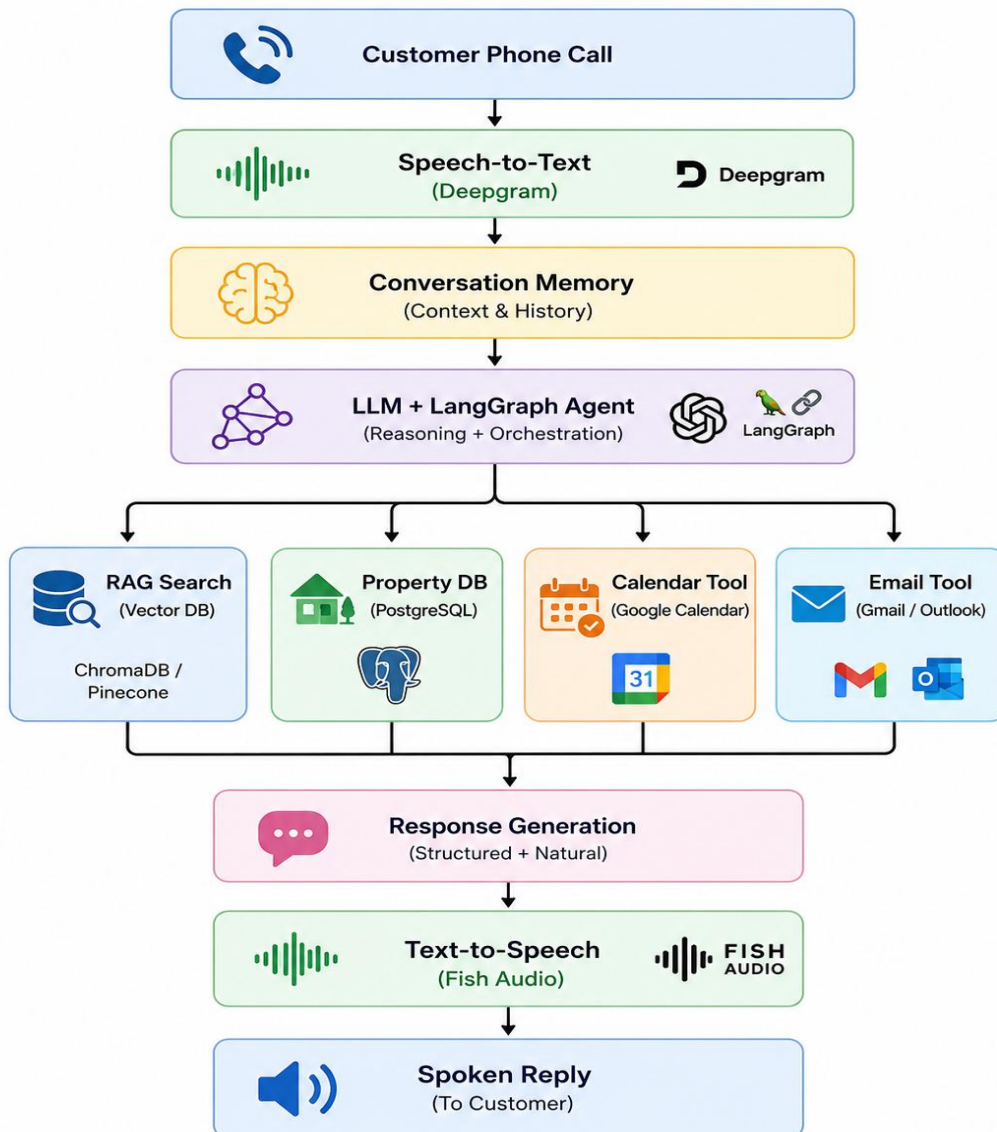
Week 7 – Day 1

Task 1: Modern AI Voice Agent Architecture

Introduction

An AI voice agent is an intelligent system that communicates with users through spoken language. It listens to speech, converts it into text, understands the user's intent, retrieves verified information, uses external tools when necessary, and responds with natural speech. In this project, the agent acts as a professional real estate sales representative capable of answering inquiries, recommending properties, scheduling appointments, and maintaining conversational context.

Architecture Diagram



Architecture Components

Speech-to-Text (STT)

Converts spoken audio into text. Recommended: Deepgram Nova-3 for low latency and streaming.

LLM Reasoning

Analyzes user intent, extracts requirements, decides whether tools or retrieval are needed, and generates responses.

Tool Calling

Allows the LLM to interact with external services such as property search, Google Calendar, email, and CRM.

Retrieval (RAG)

Retrieves verified company information from brochures, FAQs, and documents to reduce hallucinations.

Memory

Maintains conversation context such as budget, location, and appointment status.

Text-to-Speech (TTS)

Converts the generated response into natural UrduLish speech. Recommended: Fish Audio.

Telephony

Connects the AI system to phone calls and streams audio between the caller and the backend.

Workflow Orchestration

Coordinates multi-step business processes using LangGraph, such as booking appointments and sending confirmations.

End-to-End Request Flow

- Customer calls the real estate office.
- Telephony platform receives the call.
- Speech-to-Text converts speech into text.
- Conversation memory is updated.
- LLM analyzes the request and detects intent.
- LangGraph decides which tools to invoke.
- Property database/RAG/Calendar/Email tools execute.
- LLM generates a grounded response.
- Fish Audio converts text into speech.
- The customer hears the response and the conversation continues.

Suggested Technology Stack

Layer	Technology	Reason
Speech-to-Text	Deepgram Nova-3	Low-latency streaming
LLM	GPT-5.5 / Claude / Gemini	Reasoning & tool calling
Agent	LangGraph	Workflow orchestration
Vector DB	ChromaDB	RAG
Database	PostgreSQL	Structured property data
Text-to-Speech	Fish Audio	Natural UrduLish
Backend	FastAPI	API services
Scheduling	Google Calendar API	Appointments
Email	Gmail API	Notifications

Design Justification

This modular architecture separates speech processing, reasoning, retrieval, business workflows, and response generation. It reduces hallucinations through RAG, enables real-world actions through tool calling, maintains conversational context with memory, and supports scalable deployment by keeping each component independent.

Conclusion

The proposed architecture provides a production-ready foundation for a real estate AI voice agent capable of natural UrduLish conversations, grounded property recommendations, appointment management, and business workflow automation.
